

Appendix A - Hunter Marine Park Natural Values Report

Temperate East Marine Parks Network

Table of contents

Natural Values	1
Sampling summary	2
Benthic ecosystem extent	3
Observations	3
Dominant habitat types	4
Functional reef distribution and probability	5
Benchmarks of benthic natural values	6
Sediment	6
Sessile invertebrates	8
Benchmarks of demersal fish natural values	10
Survey effort	10
Species richness	11
Total abundance	13
Large Reef Fish Index*	15
Reef fish thermal index	17

Natural Values

Understanding of natural values

Sampling summary

Table 1: Summary of monitoring campaigns in the study area, showing the survey dates, sampling method, management zones sampled, and the number of samples with each data type. SPZ = Special Purpose Zone.

Date	Campaign name	Method	Areas	Samples	Habitat	Fish	Length
					Count	Count	
Aug-Sep 2025	202508 Hunter AMP stereoBRUV	BRUV	Special Purpose Zone (Trawl)	210	208	207	201

Benthic ecosystem extent

Observations

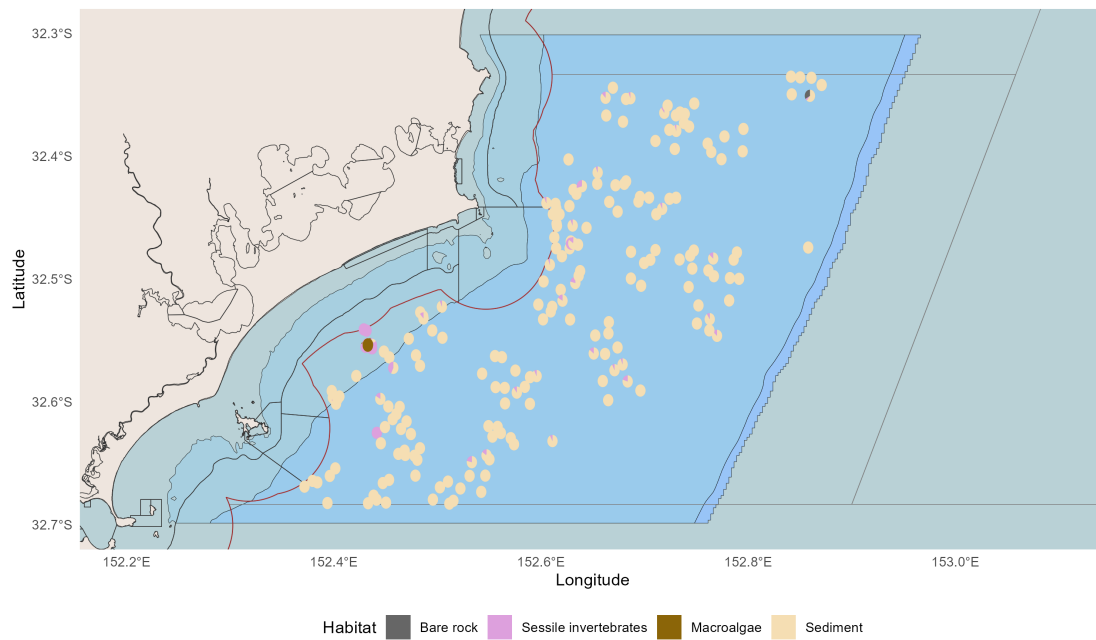


Figure 1: Habitat classes observed in spatially balanced stereo-BRUV and drop-camera imagery visualised as spatial pie charts. Commonwealth and State marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30, 70 and 200 m is shown.

Dominant habitat types

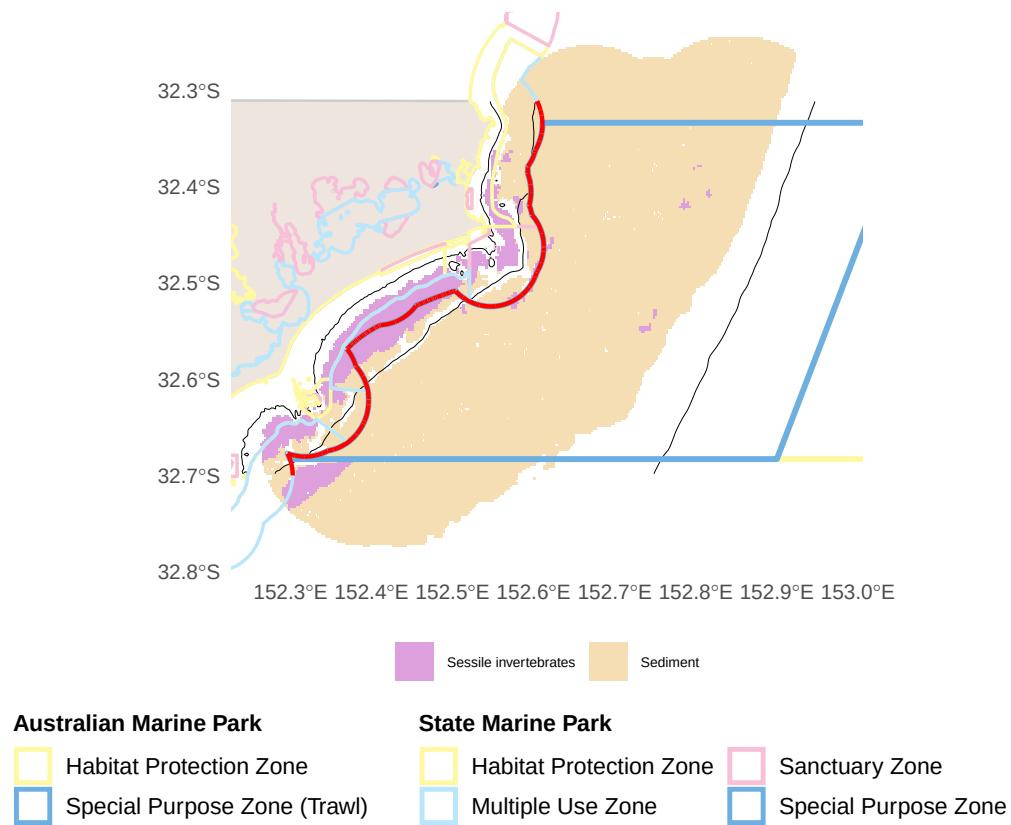


Figure 2: Predicted dominant habitat type from stereo-BRUV groundtruthing. Commonwealth and State marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30, 70 and 200 m is shown.

Functional reef distribution and probability

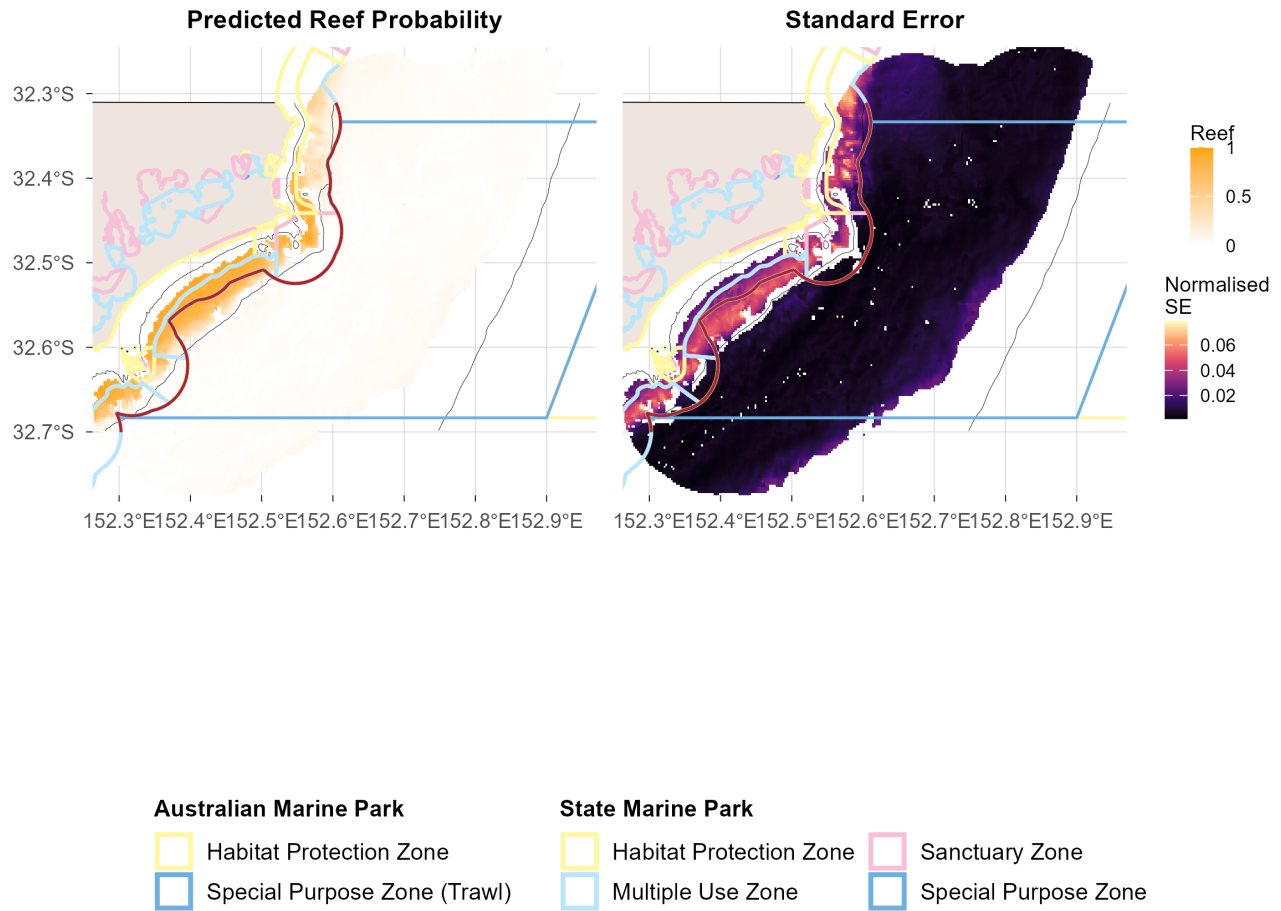


Figure 3: Predicted reef distribution probability and uncertainty from stereo-BRUV groundtruthing. Commonwealth and State marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30, 70 and 200 m is shown.

Benchmarks of benthic natural values

Sediment

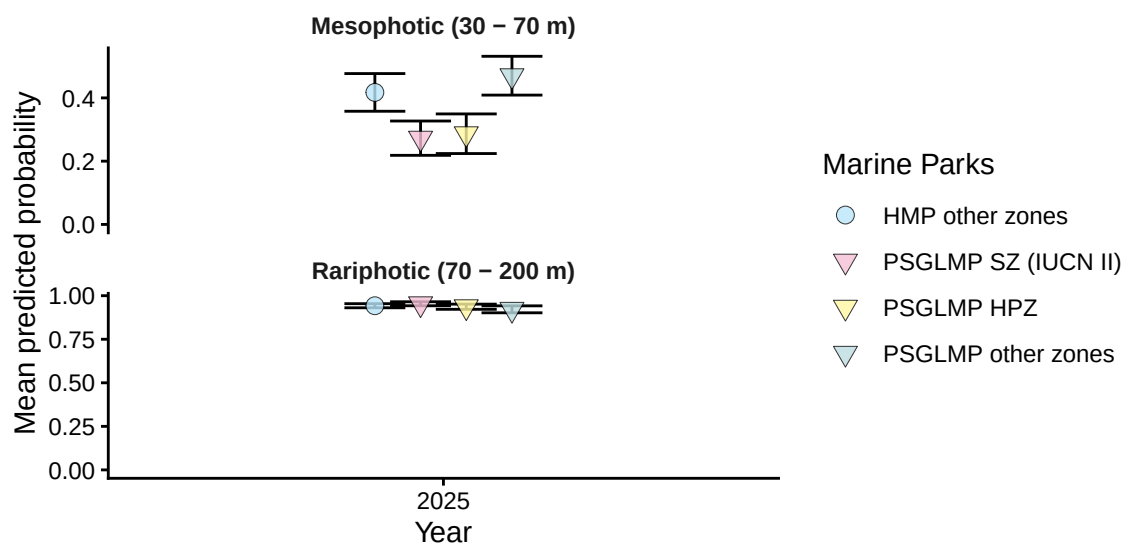


Figure 4.1: Sediment: temporal plots by depth contour from stereo-BRUV groundtruthing. HMP = Hunter Marine Park (Commonwealth), PSGLMP = Port Stephens Great Lakes Marine Park (State).

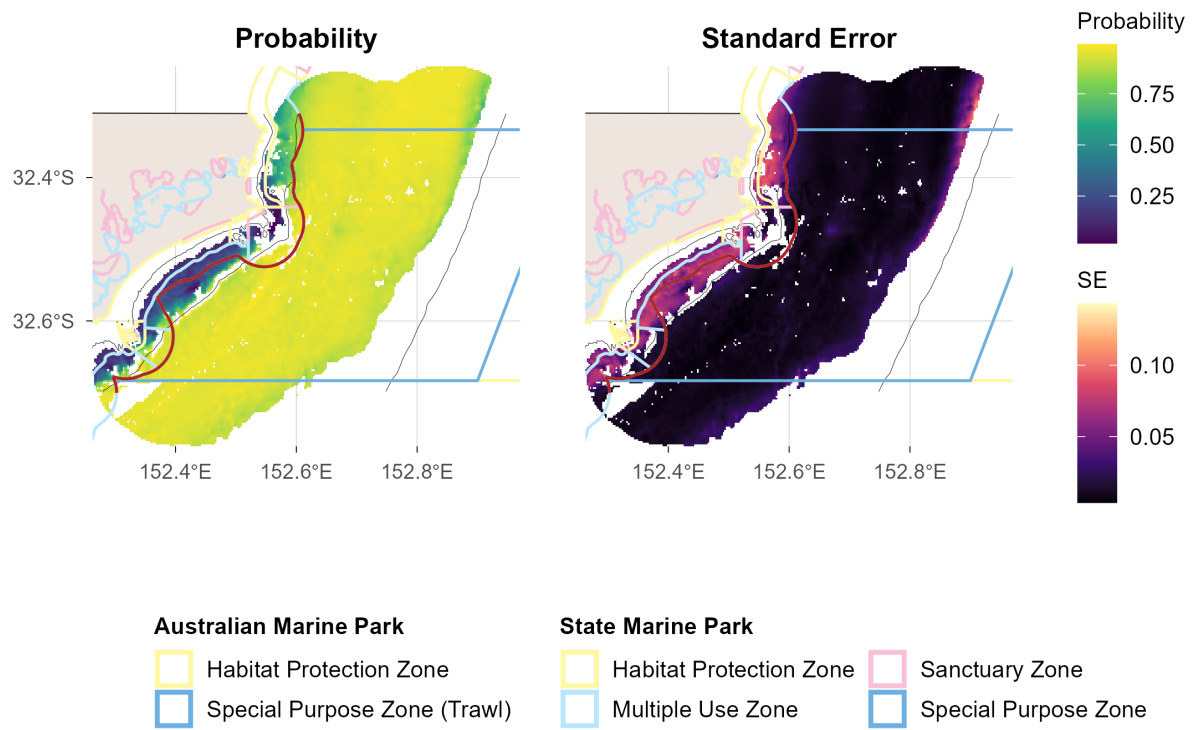


Figure 4.2: Sediment: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30, 70 and 200 m is shown.

Sessile invertebrates

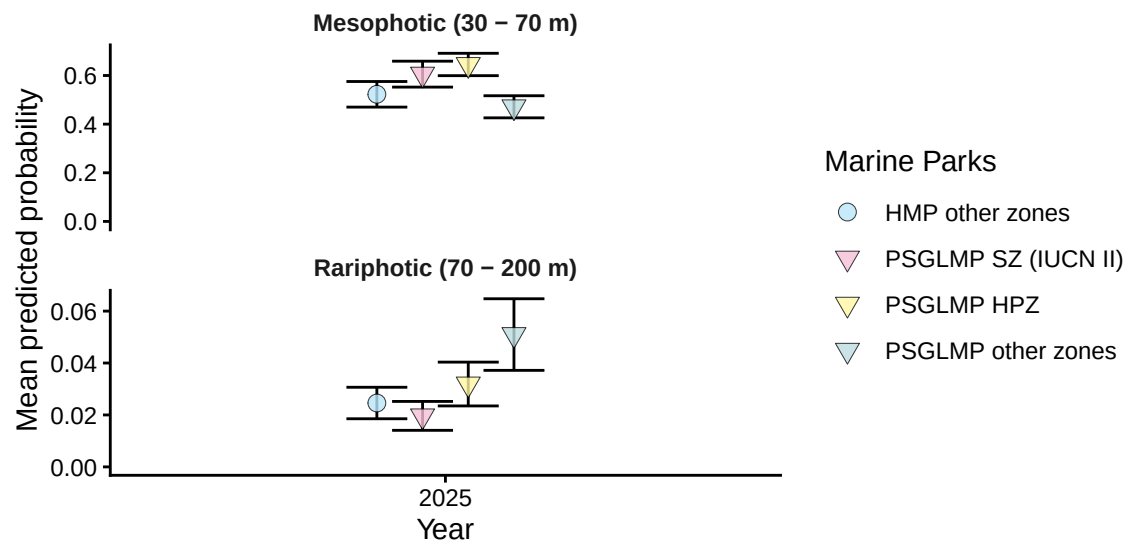


Figure 5.1: Sessile invertebrates: temporal plots by depth contour from stereo-BRUV groundtruthing. HMP = Hunter Marine Park (Commonwealth), PSGLMP = Port Stephens Great Lakes Marine Park (State).

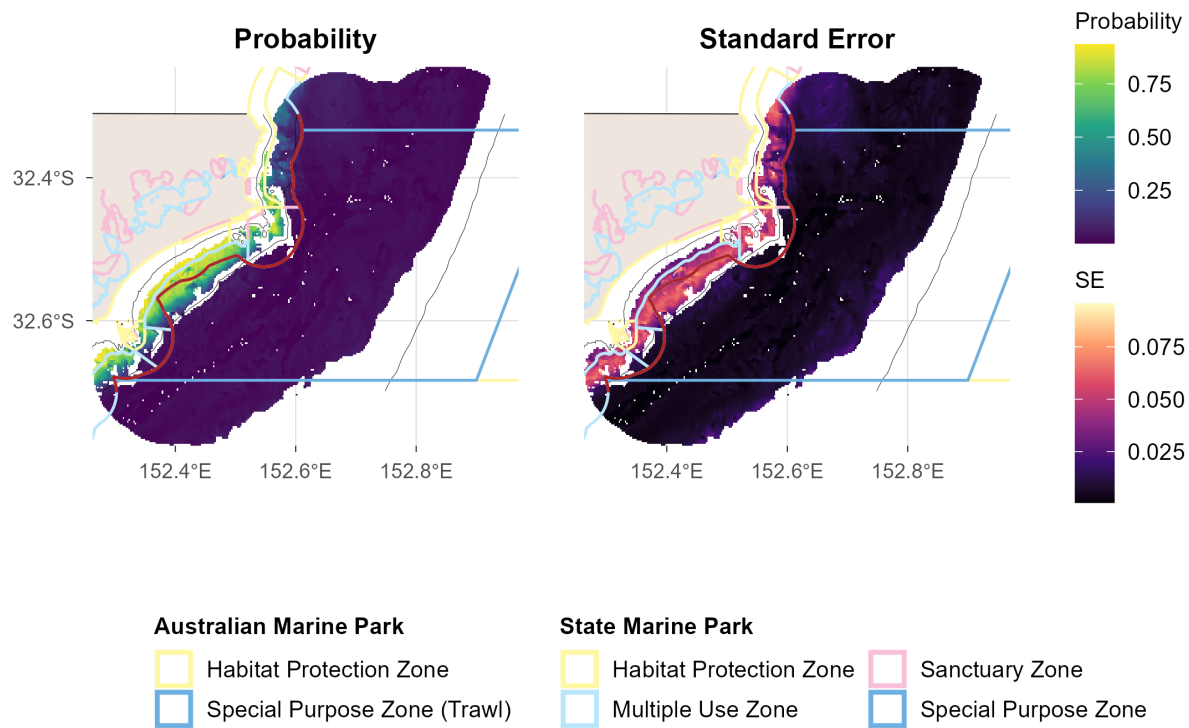


Figure 5.2: Sessile invertebrates: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. Commonwealth and State marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30, 70 and 200 m is shown.

Benchmarks of demersal fish natural values

Survey effort

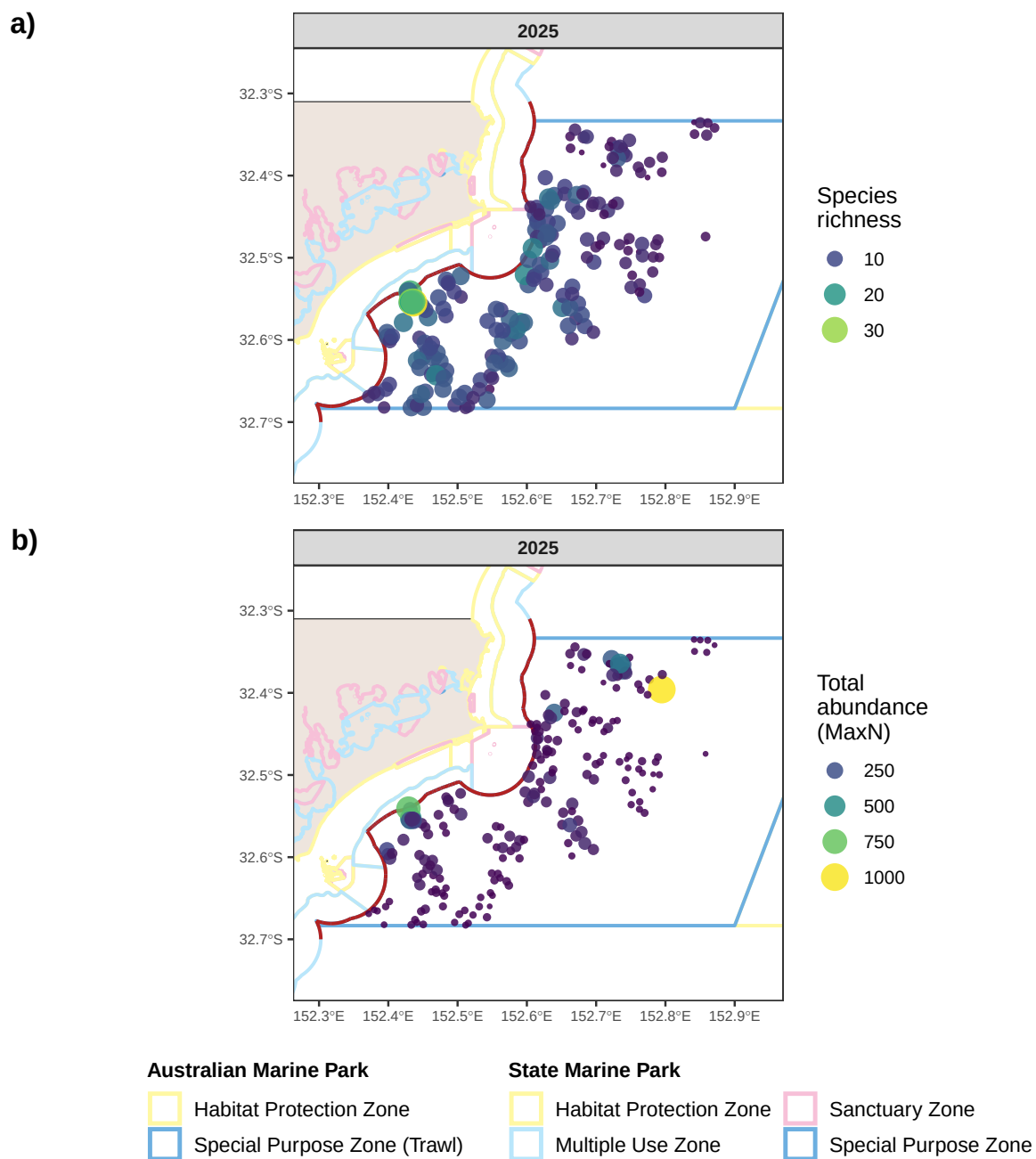


Figure 6: Spatial distribution of observed species richness and total abundance (MaxN) per stereo-BRUV drop, pooled across all deployments. Commonwealth and State marine park boundaries are shown. The red line delimits state and commonwealth waters.

Species richness

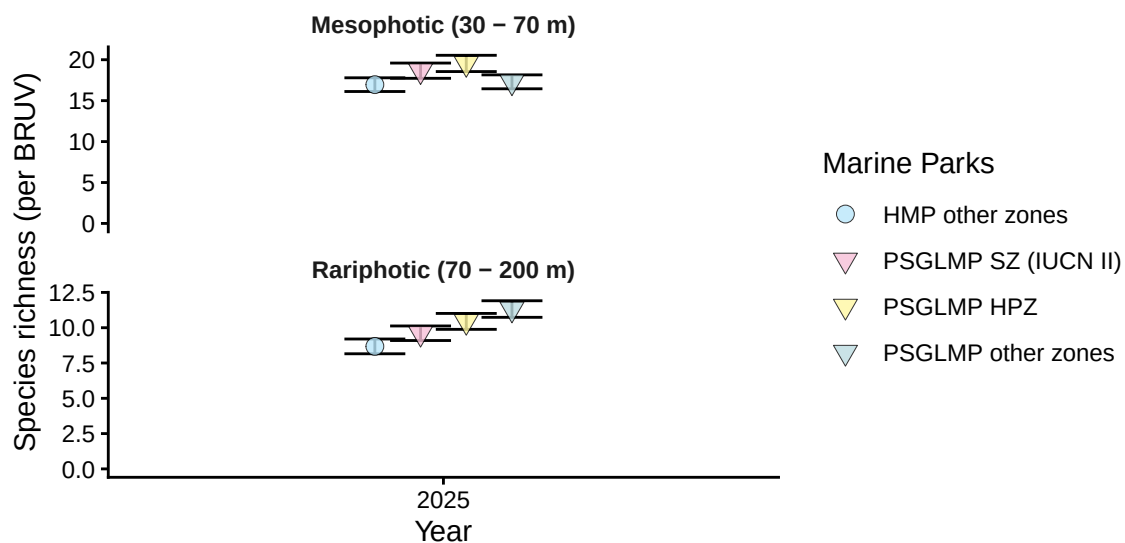


Figure 7.1: Demersal fish - Species richness: benchmark plots by zone and depth contour from stereo-BRUV groundtruthing. HMP = Hunter Marine Park (Commonwealth), PSGLMP = Port Stephens Great Lakes Marine Park (State).

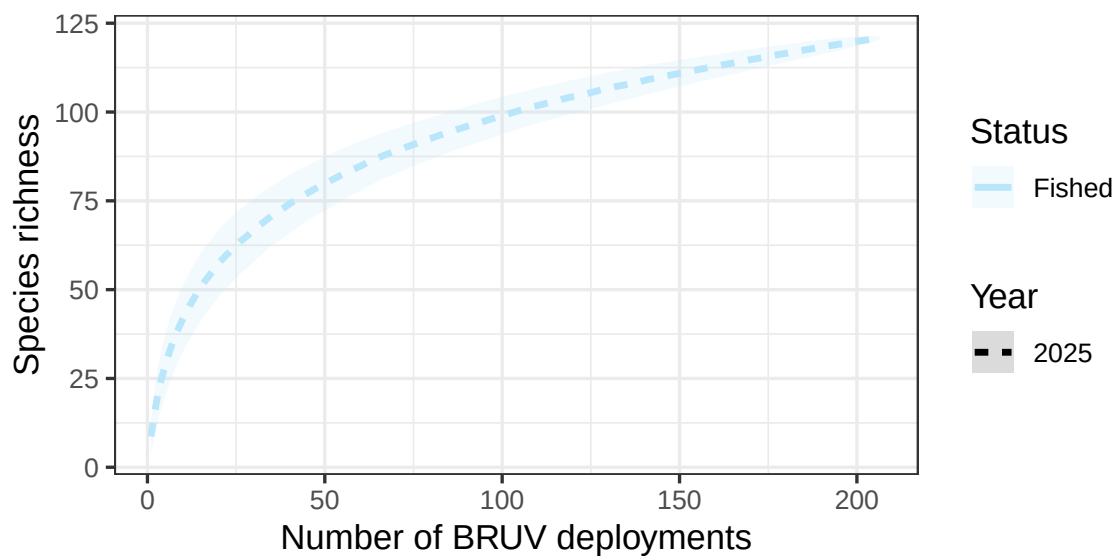


Figure 7.2: Demersal fish - Species richness: species accumulation curve by year from stereo-BRUV groundtruthing in Hunter Marine Park (Commonwealth).

Species richness

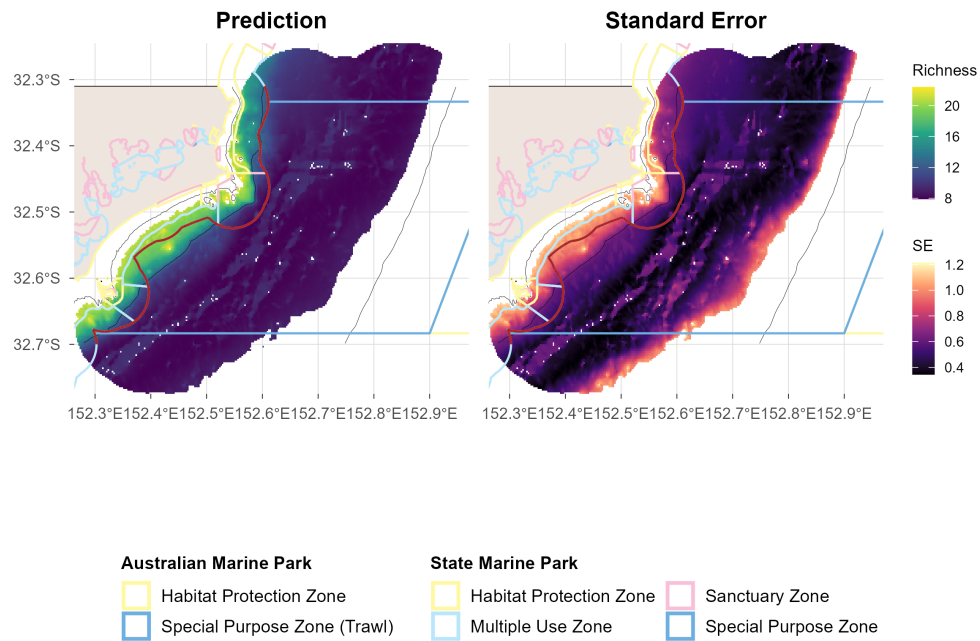


Figure 7.3: Demersal fish - Species richness: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. Commonwealth and State marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30, 70 and 200 m is shown.

Total abundance

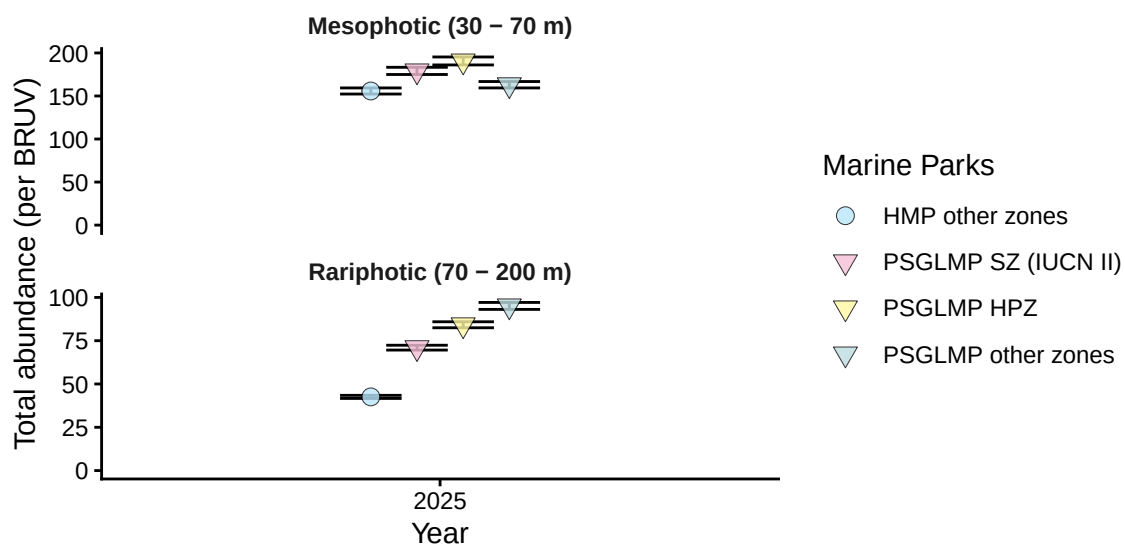


Figure 8.1: Demersal fish - Total abundance: benchmark plots by zone and depth contour from stereo-BRUV groundtruthing. HMP = Hunter Marine Park (Commonwealth), PSGLMP = Port Stephens Great Lakes Marine Park (State).

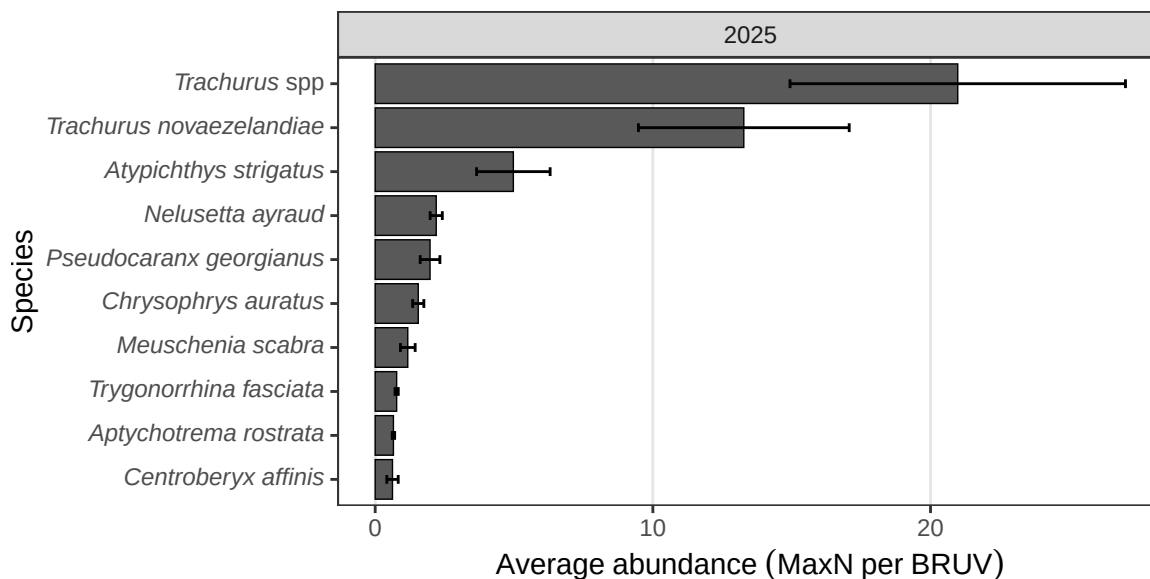


Figure 8.2: Demersal fish - Total abundance: top 10 most abundant species (pooled across deployments) from stereo-BRUV groundtruthing in Hunter Marine Park (Commonwealth).

Total abundance

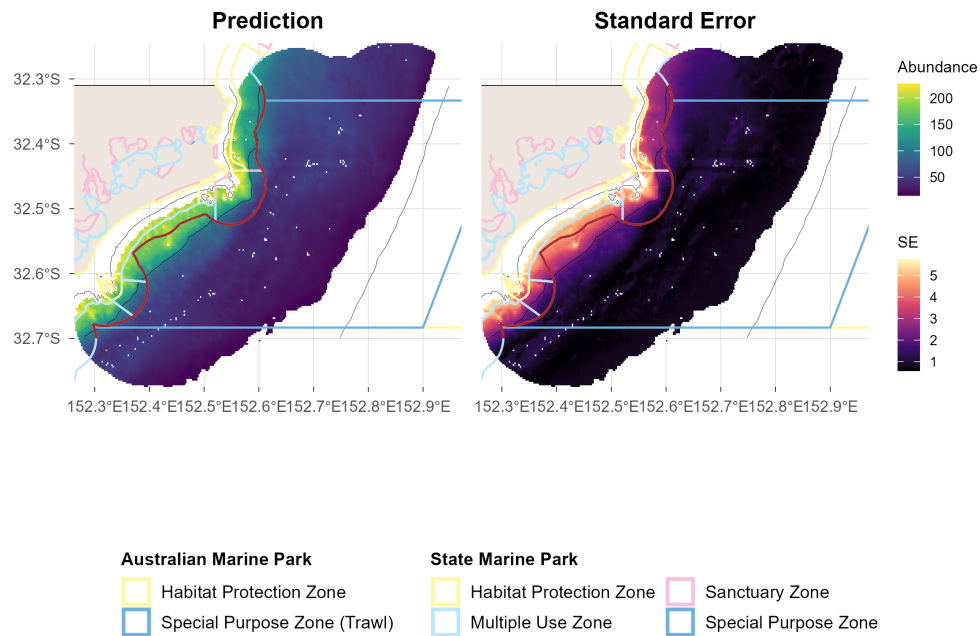


Figure 8.3: Demersal fish - Total abundance: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. Commonwealth and State marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30, 70 and 200 m is shown.

Large Reef Fish Index*

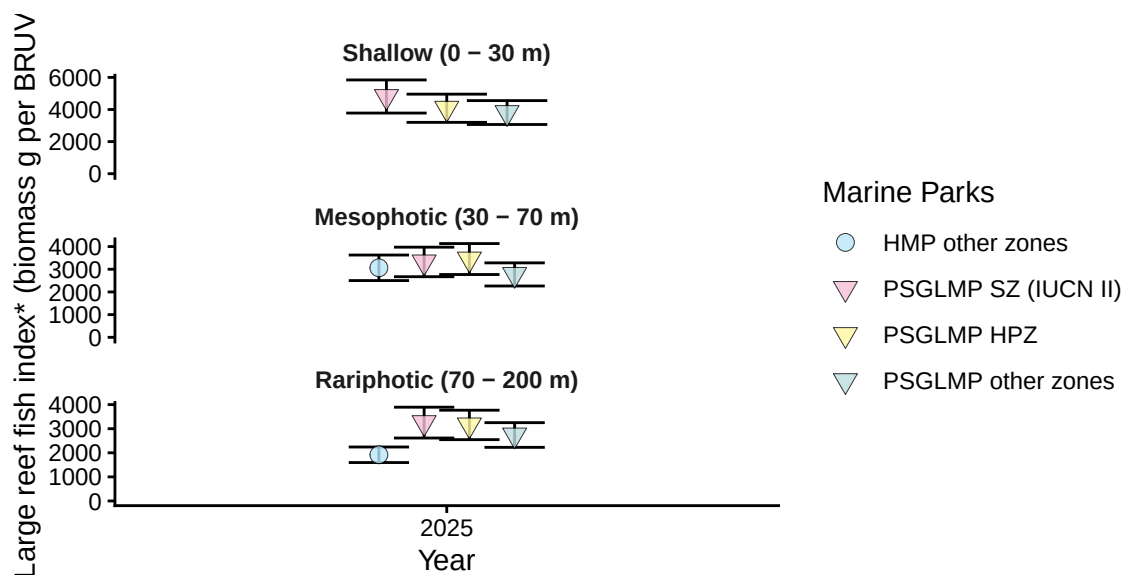


Figure 9.1: Demersal fish - Large Reef Fish Index*: benchmark plots by zone and depth contour from stereo-BRUV groundtruthing. HMP = Hunter Marine Park (Commonwealth), PSGLMP = Port Stephens Great Lakes Marine Park (State). *Large Reef Fish Index representing the biomass of all fish greater than 20 cm (B20) has been adapted here for stereo-BRUV data by removing sharks and rays.

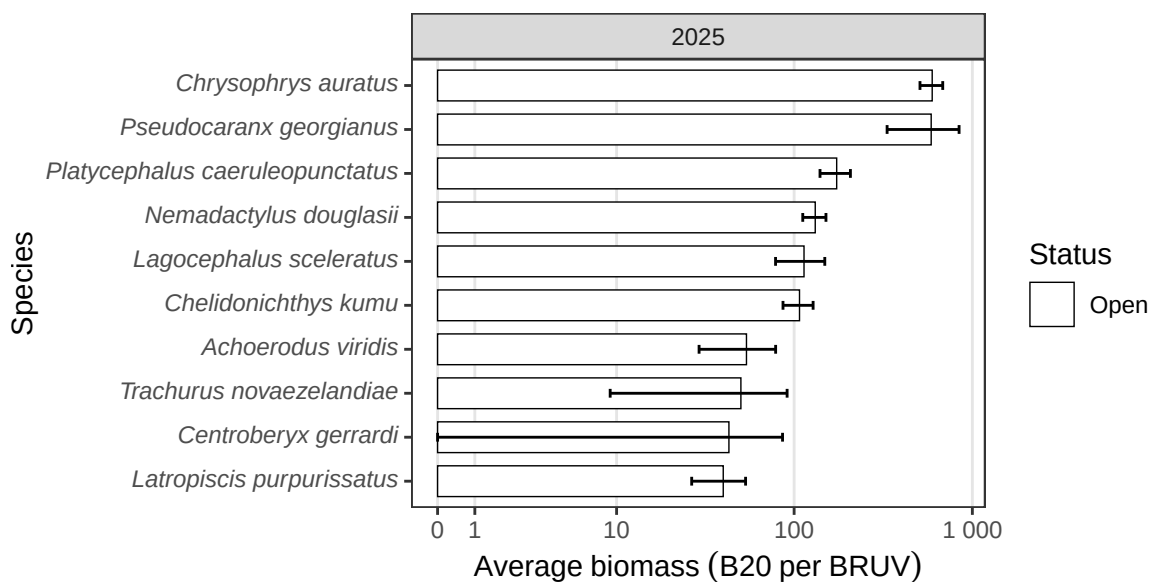


Figure 9.2: Demersal fish - Large Reef Fish Index*: top 10 greatest biomass species (pooled across deployments) from stereo-BRUV groundtruthing in Hunter Marine Park (Commonwealth).

Large Reef Fish Index*

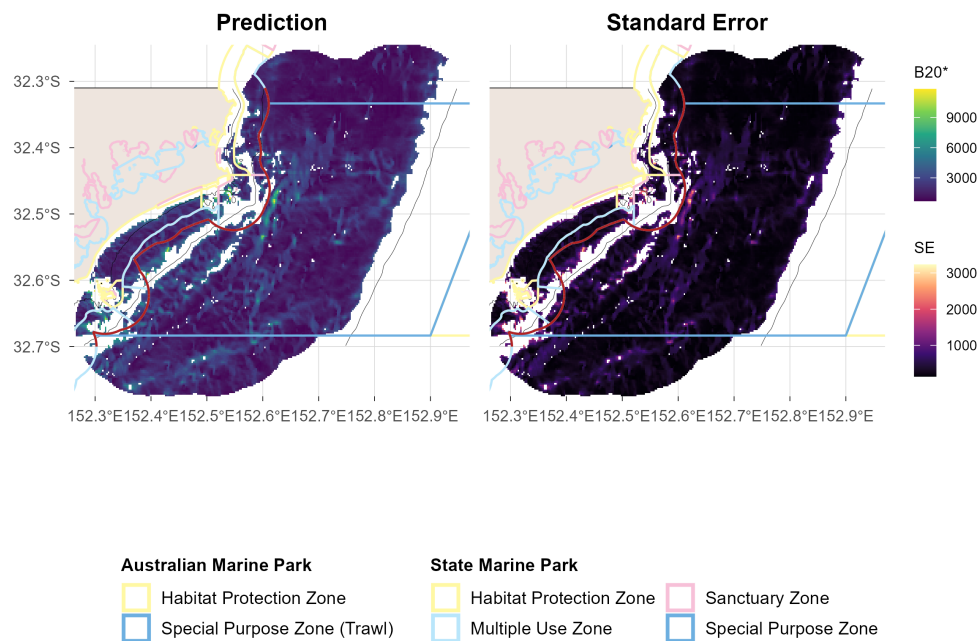


Figure 9.3: Demersal fish - Large Reef Fish Index*: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. Commonwealth and State marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30, 70 and 200 m is shown.

Reef fish thermal index

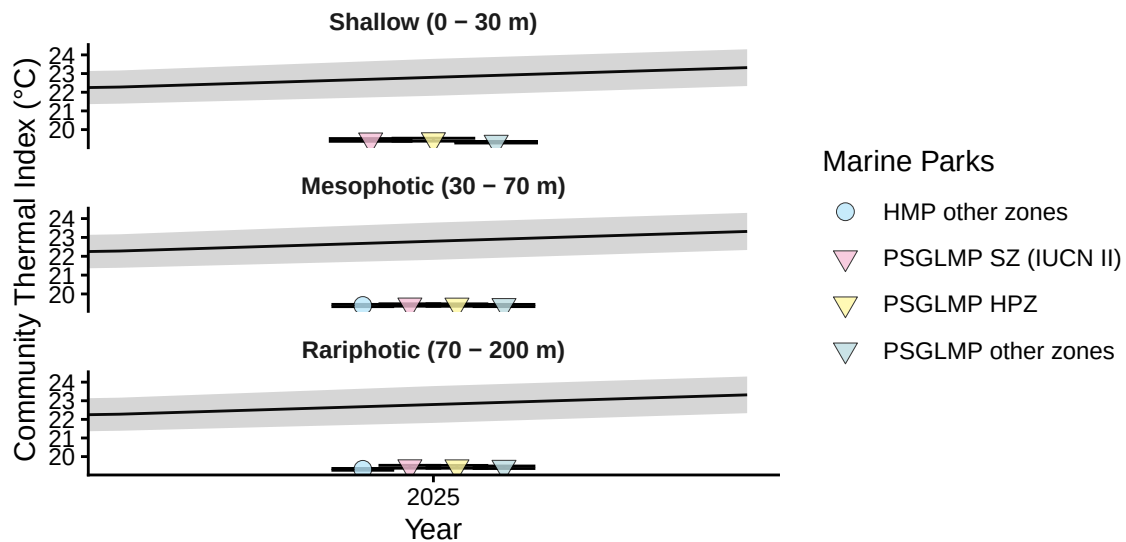


Figure 10.1: Demersal fish - Reef Fish Thermal Index: benchmark plots by zone and depth contour, with monthly sea surface temperature (black line, grey band = SD) overlaid, from stereo-BRUV groundtruthing in Hunter Marine Park (Commonwealth).

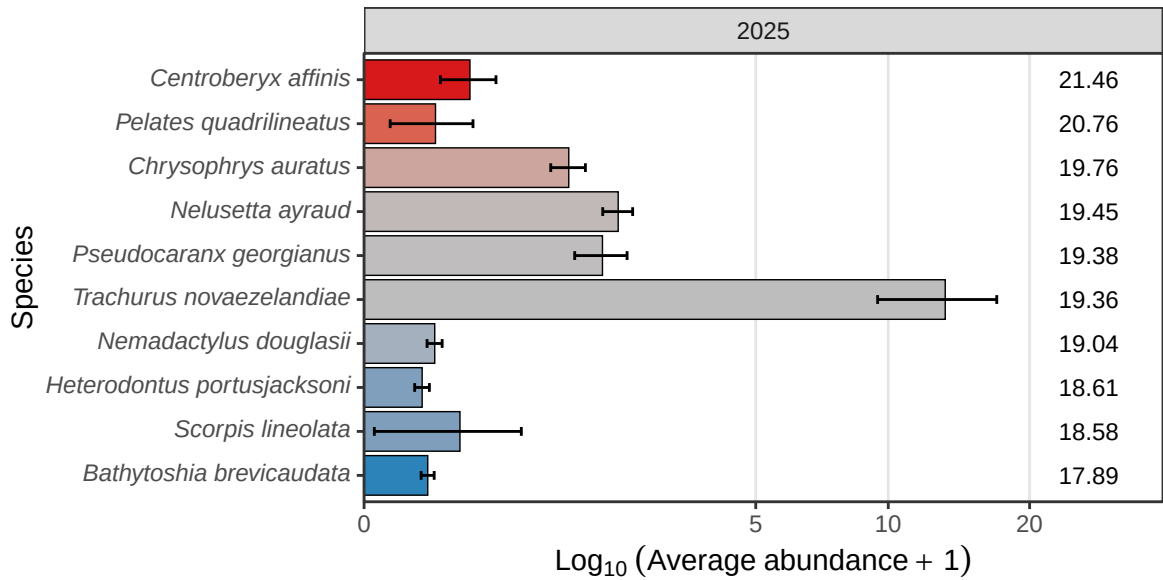


Figure 10.2: Demersal fish - Reef Fish Thermal Index: species thermal niche of top 10 most abundant species (pooled across deployments) from stereo-BRUV groundtruthing in Hunter Marine Park (Commonwealth).

Reef fish thermal index

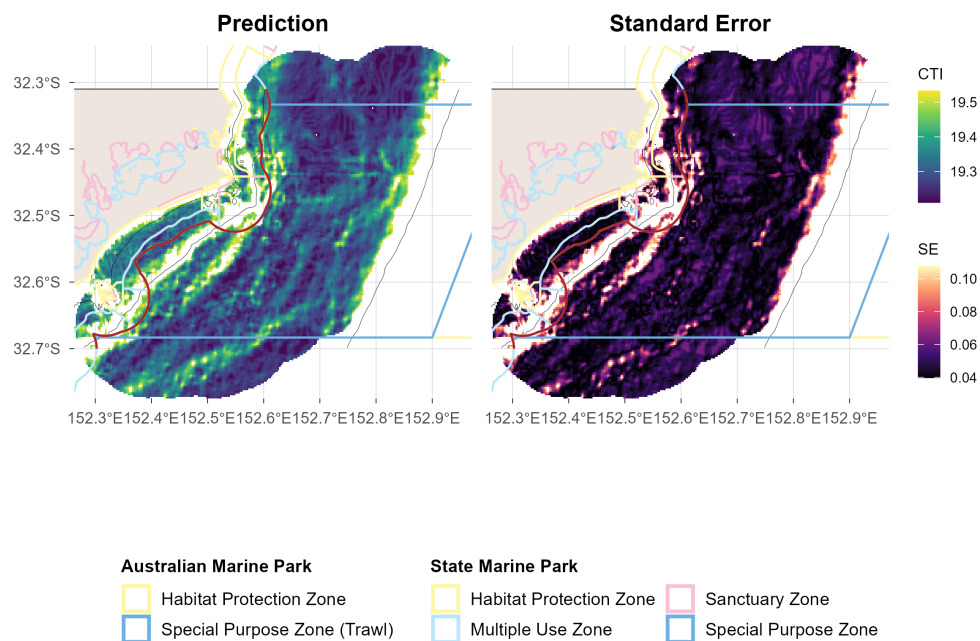


Figure 10.3: Demersal fish - Reef Fish Thermal Index: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. Commonwealth and State marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30, 70 and 200 m is shown.